

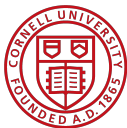
Informational and operational characterizations of dynamic quantum resources

Kaiyuan Ji¹ Eric Chitambar²

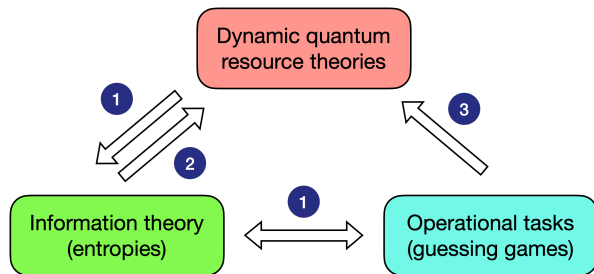
¹Cornell University

²University of Illinois at Urbana-Champaign

June 27, 2023



Outline



1. Generalizing **entropies** from quantum theory to **quantum resource theories**
2. Application: using generalized **entropies** to characterize quantum **resources**
3. Scheme: realizing **resource** characterization using **operational tasks**

Generalizing entropies from quantum theory to quantum resource theories

Using free (conditional) min-entropy to characterize quantum resources

Characterizing dynamic quantum resources with operational tasks

Discussions

What is entropy?

Two ways to interpret:

1. Degree of disorder of a system
 - ▶ An intrinsic property of the **system**

What is entropy?

Two ways to interpret:

1. Degree of disorder of a system
 - ▶ An intrinsic property of the **system**
2. An observer's degree of uncertainty (lack of knowledge) about a system
 - ▶ A property of the **system**
 - ▶ Also depends on the **observer**

What is entropy?

Two ways to interpret:

1. Degree of disorder of a system
 - ▶ An intrinsic property of the **system**
2. An observer's degree of uncertainty (lack of knowledge) about a system
 - ▶ A property of the **system**
 - ▶ Also depends on the **observer**

Normally, they are not differentiated

- because we tacitly assume the observer to be a “quantum agent”
- operationally restricted by and only by quantum mechanics

What is entropy?

Two ways to interpret:

1. Degree of disorder of a system
 - ▶ An intrinsic property of the **system**
2. An observer's degree of uncertainty (lack of knowledge) about a system
 - ▶ A property of the **system**
 - ▶ Also depends on the **observer**

Normally, they are not differentiated

- because we tacitly assume the observer to be a “quantum agent”
- operationally restricted by and only by quantum mechanics

However, the observer could be more general and flexible

- capable of a **subset** O_{free} of quantum operations, called **free operations**
- We want the freedom of choosing O_{free} according to our needs

Uncertainty in quantum resource theories

Free entropy: Degree of uncertainty (lack of knowledge) about a system from an **operationally restricted** observer's perspective

- depends on both the **system** and the **free operations** of the observer
- generalizes quantum entropy

Uncertainty in quantum resource theories

Free entropy: Degree of uncertainty (lack of knowledge) about a system from an **operationally restricted** observer's perspective

- depends on both the **system** and the **free operations** of the observer
- generalizes quantum entropy

If the set of free operations $\mathcal{O}_{\text{free}}$ is physically motivated

- contains the identity channel
- closed under composition
- is (topologically) closed and convex

Uncertainty in quantum resource theories

Free entropy: Degree of uncertainty (lack of knowledge) about a system from an **operationally restricted** observer's perspective

- depends on both the **system** and the **free operations** of the observer
- generalizes quantum entropy

If the set of free operations $\mathcal{O}_{\text{free}}$ is physically motivated

- contains the identity channel
- closed under composition
- is (topologically) closed and convex

Then it specifies a (convex) **quantum resource theory** ($\mathcal{O}_{\text{free}}, \mathcal{D}_{\text{free}}$)¹

- $\mathcal{D}_{\text{free}}$ is the set of states producible using free operations, called **free states**

¹E. Chitambar and G. Gour, Rev. Mod. Phys. 91, 025001 (2019)

Uncertainty in quantum resource theories

Free entropy: Degree of uncertainty (lack of knowledge) about a system from an **operationally restricted** observer's perspective

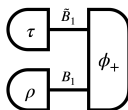
- depends on both the **system** and the **free operations** of the observer
- generalizes quantum entropy

Free entropy is a resource-theory generalization of quantum entropy

Free min-entropy

Quantum min-entropy:

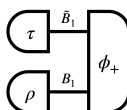
$$H_{\min}(B_1)_\rho := -\log \min \left\{ r \mid \rho_{B_1} - r \rho_{B_1} \geq 0 \right\}$$

$$= -\log \max_{\tau \in \mathcal{D}^{B_1}} d_{B_1}$$


Free min-entropy

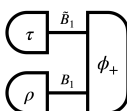
Quantum min-entropy:

$$H_{\min}(B_1)_\rho := -\log \min \left\{ r : r I^{B_1} - \rho^{B_1} \geq 0 \right\}$$

$$= -\log \max_{\tau \in \mathcal{D}^{B_1}} d_{B_1}$$


Free min-entropy:

$$H_{\min}^{\text{free}}(B_1)_\rho := -\log \min \left\{ r : r I^{B_1} - \rho^{B_1} \in \text{cone}^*(\mathcal{D}_{\text{free}}^{B_1}) \right\}$$

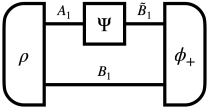
$$= -\log \max_{\tau \in \mathcal{D}_{\text{free}}^{B_1}} d_{B_1}$$


What we did: replacing quantum states $\mathcal{D}^{B_1}$ with **free states** $\mathcal{D}_{\text{free}}^{B_1}$

Free conditional min-entropy

Quantum conditional min-entropy:

$$H_{\min}(B_1|A_1)_\rho := -\log \min \left\{ \gamma^{A_1} \otimes I^{B_1} - \rho^{A_1 B_1} \geq 0 \right\}$$

$$= -\log \max_{\Psi \in C^{A_1 \rightarrow B_1}} d_{B_1}$$


Free conditional min-entropy

Quantum conditional min-entropy:

$$H_{\min}(B_1|A_1)_\rho := -\log \min \left\{ \text{Tr}[\gamma] : \gamma^{A_1} \otimes I^{B_1} - \rho^{A_1 B_1} \geq 0 \right\}$$

$$= -\log \max_{\Psi \in C^{A_1 \rightarrow B_1}} d_{B_1} \left(\begin{array}{c} \text{Diagram: A box labeled } \rho \text{ on the left and } \phi_+ \text{ on the right. A top wire labeled } A_1 \text{ goes from } \rho \text{ to a box labeled } \Psi. \text{ A bottom wire labeled } B_1 \text{ goes from } \rho \text{ to } \phi_+. \text{ A wire labeled } \tilde{B}_1 \text{ goes from } \Psi \text{ to } \phi_+. \end{array} \right)$$

Free conditional min-entropy:

$$H_{\min}^{\text{free}}(B_1|A_1)_\rho := -\log \min \left\{ \text{Tr}[\gamma] : \gamma^{A_1} \otimes I^{B_1} - \rho^{A_1 B_1} \in \text{cone}^*(O_{\text{free}}^{A_1 \rightarrow B_1}) \right\}$$

$$= -\log \max_{\Psi \in O_{\text{free}}^{A_1 \rightarrow B_1}} d_{B_1} \left(\begin{array}{c} \text{Diagram: A box labeled } \rho \text{ on the left and } \phi_+ \text{ on the right. A top wire labeled } A_1 \text{ goes from } \rho \text{ to a box labeled } \Psi. \text{ A bottom wire labeled } B_1 \text{ goes from } \rho \text{ to } \phi_+. \text{ A wire labeled } \tilde{B}_1 \text{ goes from } \Psi \text{ to } \phi_+. \end{array} \right)$$

What we did: replacing quantum channels $C^{A_1 \rightarrow B_1}$ with **free operations**

$O_{\text{free}}^{A_1 \rightarrow B_1}$

Free conditional min-entropy

Observation:

- ▶ Free (conditional) min-entropy $\geq$ Quantum (conditional) min-entropy

Free conditional min-entropy

Observation:

- ▶ Free (conditional) min-entropy $\geq$ Quantum (conditional) min-entropy
 - ▶ Operational restriction debilitates your understanding of information
 - ▶ You become more uncertain about a given system

Free conditional min-entropy

Observation:

- ▶ Free (conditional) min-entropy $\geq$ Quantum (conditional) min-entropy
 - ▶ Operational restriction debilitates your understanding of information
 - ▶ You become more uncertain about a given system

Example (Classical agent observes qubit system)

Free states: incoherent (i.e., diagonal) states.

- ▶ $H_{\min}(B_1)_{|+\rangle\langle+|} = 0$, $H_{\min}(B_1)_{\frac{I}{2}} = 1$
- ▶ $H_{\min}^{\text{free}}(B_1)_{|+\rangle\langle+|} = H_{\min}^{\text{free}}(B_1)_{\frac{I}{2}} = 1$

Explanation:

- ▶ $H_{\min}(B_1)_{|+\rangle\langle+|}$: the “intrinsic” disorder of $|+\rangle\langle+|$, which is 0
- ▶ $H_{\min}^{\text{free}}(B_1)_{|+\rangle\langle+|} - H_{\min}(B_1)_{|+\rangle\langle+|}$: the additional uncertainty caused by the observer’s limited comprehension
- ▶ So $|+\rangle\langle+|$ **appears** equally uncertain as $\frac{I}{2}$ to the classical observer, who is unable to understand coherent information

Monotonicity

- ▶ Monotonicity under free operations on the conditioned system:

$$H_{\min}^{\text{free}}(B_1|A'_1)_{(\Psi \otimes \text{id})[\rho]} \geq H_{\min}^{\text{free}}(B_1|A_1)_{\rho}$$

if $\Psi^{A_1 \rightarrow A'_1}$ is a free operation.

Monotonicity

- ▶ Monotonicity under free operations on the conditioned system:

$$H_{\min}^{\text{free}}(B_1|A'_1)_{(\Psi \otimes \text{id})[\rho]} \geq H_{\min}^{\text{free}}(B_1|A_1)_\rho$$

if $\Psi^{A_1 \rightarrow A'_1}$ is a free operation.

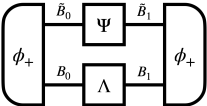
- ▶ An interesting and relevant sequence of inequalities:

$$H_{\min}(B_1)_\omega \leq H_{\min}^{\text{free}}(B_1|A_1)_{\rho \otimes \omega} \leq H_{\min}^{\text{free}}(B_1)_\omega$$

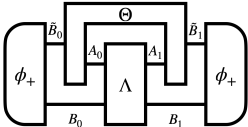
for ρ^{A_1} and ω^{B_1} .

- ▶ The more resourceful ρ is, the closer $H_{\min}^{\text{free}}(B_1|A_1)_{\rho \otimes \omega}$ is to the left
- ▶ Access to (independent) static resource helps you overcome operational restriction, and thus eliminate the uncertainty caused by “limited comprehension”

Free (conditional) min-entropy of channels

$$H_{\min}^{\text{free}}(B \rightarrow)_{\Lambda} = -\log \max_{\Psi \in \mathcal{C}_{\text{free}}^{B_0 \rightarrow B_1}} d_{B_1}$$


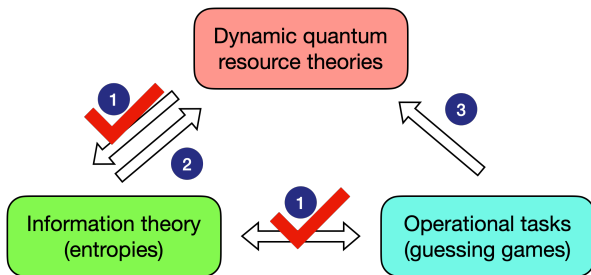
The diagram shows a quantum circuit. On the left, a large rounded rectangle contains two ancilla states, ϕ_+ . A line from the top ϕ_+ goes to a box labeled Ψ , which has two inputs: $\tilde{B}_0$ and B_0 . The output of Ψ is $\tilde{B}_1$. Below Ψ is a box labeled Λ , which has input B_0 and output B_1 . The output B_1 goes to another ϕ_+ state on the right.

$$H_{\min}^{\text{free}}(B \rightarrow | A \rightarrow)_{\Lambda} = -\log \max_{\Theta \in \mathcal{S}_{\text{free}}^{(A \rightarrow) \rightarrow (B \rightarrow)}} d_{B_1}$$


The diagram shows a quantum circuit. On the left, a large rounded rectangle contains two ancilla states, ϕ_+ . A line from the top ϕ_+ goes to a box labeled Θ , which has three inputs: $\tilde{B}_0$, A_0 , and B_0 . The output of Θ is $\tilde{B}_1$. Below Θ is a box labeled Λ , which has input B_0 and output B_1 . The output B_1 goes to another ϕ_+ state on the right. There is also an output A_1 from the Λ box.

- Reduce to the extended quantum (conditional) min-entropy of channels if all quantum superchannels are free superoperations¹

¹G. Gour, IEEE Transactions on Information Theory 65, 5880 (2019).



Generalizing entropies from quantum theory to quantum resource theories

Using free (conditional) min-entropy to characterize quantum resources

Characterizing dynamic quantum resources with operational tasks

Discussions

Constructing resource monotones from monotonicity of free conditional min-entropy

Definition (Resource monotone ²)

A function $f: \bigcup_{A_1} \mathcal{D}^{A_1} \rightarrow \mathbb{R}$ satisfying

$$f(\Psi[\rho]) \leq f(\rho)$$

for any free operation $\Psi^{A_1 \rightarrow A'_1}$.

²E. Chitambar and G. Gour, Rev. Mod. Phys. 91, 025001 (2019)

Constructing resource monotones from monotonicity of free conditional min-entropy

Definition (Resource monotone ²)

A function $f: \bigcup_{A_1} \mathcal{D}^{A_1} \rightarrow \mathbb{R}$ satisfying

$$f(\Psi[\rho]) \leq f(\rho)$$

for any free operation $\Psi^{A_1 \rightarrow A'_1}$.

- Recall: monotonicity of free conditional min-entropy:

$$H_{\min}^{\text{free}}(B_1|A'_1)_{(\Psi \otimes \text{id})[\rho]} \geq H_{\min}^{\text{free}}(B_1|A_1)_{\rho}$$

if $\Psi^{A_1 \rightarrow A'_1}$ is a free operation.

²E. Chitambar and G. Gour, Rev. Mod. Phys. 91, 025001 (2019)

Constructing resource monotones from monotonicity of free conditional min-entropy

Definition (Resource monotone ²)

A function $f: \bigcup_{A_1} \mathcal{D}^{A_1} \rightarrow \mathbb{R}$ satisfying

$$f(\Psi[\rho]) \leq f(\rho)$$

for any free operation $\Psi^{A_1 \rightarrow A'_1}$.

- Recall: monotonicity of free conditional min-entropy:

$$H_{\min}^{\text{free}}(B_1|A'_1)_{(\Psi \otimes \text{id})[\rho]} \geq H_{\min}^{\text{free}}(B_1|A_1)_{\rho}$$

if $\Psi^{A_1 \rightarrow A'_1}$ is a free operation.

- The function $f_{\omega}: \rho^{A_1} \mapsto -H_{\min}^{\text{free}}(B_1|A_1)_{\rho \otimes \omega}$ is a resource monotone for any given ω^{B_1} .

²E. Chitambar and G. Gour, Rev. Mod. Phys. 91, 025001 (2019)

Entropic conditions for free convertibility

Every resource monotone f provides a necessary condition for free convertibility (by definition)

- ▶ If $\rho \succcurlyeq_{\text{free}} \sigma$, then $f(\rho) \geq f(\sigma)$
 - $\rho \succcurlyeq_{\text{free}} \sigma$ means $\Psi[\rho] = \sigma$ for some free operation Ψ

Entropic conditions for free convertibility

Every resource monotone f provides a necessary condition for free convertibility (by definition)

- ▶ If $\rho \succcurlyeq_{\text{free}} \sigma$, then $f(\rho) \geq f(\sigma)$
 - $\rho \succcurlyeq_{\text{free}} \sigma$ means $\Psi[\rho] = \sigma$ for some free operation Ψ
- ▶ The converse (sufficiency) is typically not true

Entropic conditions for free convertibility

Every resource monotone f provides a necessary condition for free convertibility (by definition)

- ▶ If $\rho \succcurlyeq_{\text{free}} \sigma$, then $f(\rho) \geq f(\sigma)$
 - $\rho \succcurlyeq_{\text{free}} \sigma$ means $\Psi[\rho] = \sigma$ for some free operation Ψ
- ▶ The converse (sufficiency) is typically not true

A “complete” set of resource monotones $\{f_i\}_i$:

- ▶ Each f_i provides a necessary condition for free convertibility
- ▶ They jointly provide a sufficient condition:
 - If $f_i(\rho) \geq f_i(\sigma)$ for all i , then $\rho \succcurlyeq_{\text{free}} \sigma$

Entropic conditions for free convertibility

Theorem (Entropic conditions for free convertibility between states)

For any states ρ^{A_1} and σ^{B_1} ,

$$\rho \succcurlyeq_{\text{free}} \sigma \iff H_{\min}^{\text{free}}(B'_1|A_1)_{\rho \otimes \omega} \leq H_{\min}^{\text{free}}(B'_1|B_1)_{\sigma \otimes \omega} \quad \forall \omega^{B'_1} \quad (B'_1 \cong B_1).$$

³R. Takagi and B. Regula, Phys. Rev. X 9, 031053 (2019).

⁴G. Gour and C. M. Scandolo, arXiv:2101.01552 (2021).

Entropic conditions for free convertibility

Theorem (Entropic conditions for free convertibility between states)

For any states ρ^{A_1} and σ^{B_1} ,

$$\rho \succsim_{\text{free}} \sigma \iff H_{\min}^{\text{free}}(B'_1|A_1)_{\rho \otimes \omega} \leq H_{\min}^{\text{free}}(B'_1|B_1)_{\sigma \otimes \omega} \quad \forall \omega^{B'_1} \quad (B'_1 \cong B_1).$$

Theorem (Entropic conditions for free convertibility between channels)

For any channels $\Lambda^{A_0 \rightarrow A_1}$ and $\Psi^{B_0 \rightarrow B_1}$,

$$\Lambda \succsim_{\text{free}} \Psi \iff H_{\min}^{\text{free}}(B'_{\rightarrow}|A_{\rightarrow})_{\Lambda \otimes \Omega} \leq H_{\min}^{\text{free}}(B'_{\rightarrow}|B_{\rightarrow})_{\Psi \otimes \Omega} \quad \forall \Omega^{B'_0 \rightarrow B'_1},$$

$(B'_0 \cong B_0 \text{ and } B'_1 \cong B_1).$

³R. Takagi and B. Regula, Phys. Rev. X 9, 031053 (2019).

⁴G. Gour and C. M. Scandolo, arXiv:2101.01552 (2021).

Entropic conditions for free convertibility

Theorem (Entropic conditions for free convertibility between states)

For any states ρ^{A_1} and σ^{B_1} ,

$$\rho \succsim_{\text{free}} \sigma \iff H_{\min}^{\text{free}}(B'_1|A_1)_{\rho \otimes \omega} \leq H_{\min}^{\text{free}}(B'_1|B_1)_{\sigma \otimes \omega} \quad \forall \omega^{B'_1} \quad (B'_1 \cong B_1).$$

Theorem (Entropic conditions for free convertibility between channels)

For any channels $\Lambda^{A_0 \rightarrow A_1}$ and $\Psi^{B_0 \rightarrow B_1}$,

$$\Lambda \succsim_{\text{free}} \Psi \iff H_{\min}^{\text{free}}(B'_{\rightarrow}|A_{\rightarrow})_{\Lambda \otimes \Omega} \leq H_{\min}^{\text{free}}(B'_{\rightarrow}|B_{\rightarrow})_{\Psi \otimes \Omega} \quad \forall \Omega^{B'_0 \rightarrow B'_1},$$

$(B'_0 \cong B_0 \text{ and } B'_1 \cong B_1).$

- ▶ Similar mathematical results were derived before^{3 4}
- ▶ We provide the first information-theoretic interpretation for these conditions in terms of generalized entropies

³R. Takagi and B. Regula, Phys. Rev. X 9, 031053 (2019).

⁴G. Gour and C. M. Scandolo, arXiv:2101.01552 (2021).

Entropic conditions for free convertibility

Theorem (Entropic conditions for free convertibility between states)

For any states ρ^{A_1} and σ^{B_1} ,

$$\rho \succsim_{\text{free}} \sigma \iff H_{\min}^{\text{free}}(B'_1|A_1)_{\rho \otimes \omega} \leq H_{\min}^{\text{free}}(B'_1|B_1)_{\sigma \otimes \omega} \quad \forall \omega^{B'_1} \quad (B'_1 \cong B_1).$$

Theorem (Entropic conditions for free convertibility between channels)

For any channels $\Lambda^{A_0 \rightarrow A_1}$ and $\Psi^{B_0 \rightarrow B_1}$,

$$\Lambda \succsim_{\text{free}} \Psi \iff H_{\min}^{\text{free}}(B'_{\rightarrow}|A_{\rightarrow})_{\Lambda \otimes \Omega} \leq H_{\min}^{\text{free}}(B'_{\rightarrow}|B_{\rightarrow})_{\Psi \otimes \Omega} \quad \forall \Omega^{B'_0 \rightarrow B'_1},$$

$(B'_0 \cong B_0 \text{ and } B'_1 \cong B_1).$

- ▶ Similar mathematical results were derived before^{3 4}
- ▶ We provide the first information-theoretic interpretation for these conditions in terms of generalized entropies
- ▶ The conditions can be simplified
 - reducing the size; formulated as a single conic program

³R. Takagi and B. Regula, Phys. Rev. X 9, 031053 (2019).

⁴G. Gour and C. M. Scandolo, arXiv:2101.01552 (2021).

Resource robustness as a reflection of informativeness

- Recall: $H_{\min}(B_1)_\omega \leq H_{\min}^{\text{free}}(B_1|A_1)_{\rho \otimes \omega} \leq H_{\min}^{\text{free}}(B_1)_\omega$ for ρ^{A_1} and ω^{B_1} .

Resource robustness as a reflection of informativeness

- ▶ Recall: $H_{\min}(B_1)_\omega \leq H_{\min}^{\text{free}}(B_1|A_1)_{\rho \otimes \omega} \leq H_{\min}^{\text{free}}(B_1)_\omega$ for ρ^{A_1} and ω^{B_1} .
- ▶ By monotonicity, $H_{\min}^{\text{free}}(B_1)_\omega - H_{\min}^{\text{free}}(B_1|A_1)_{\rho \otimes \omega}$, as a function of ρ^{A_1} , is a **resource measure** for any ω^{B_1} .
 - non-increasing under free operations
 - equal to 0 if ρ^{A_1} is a free state

Resource robustness as a reflection of informativeness

- ▶ Recall: $H_{\min}(B_1)_\omega \leq H_{\min}^{\text{free}}(B_1|A_1)_{\rho \otimes \omega} \leq H_{\min}^{\text{free}}(B_1)_\omega$ for ρ^{A_1} and ω^{B_1} .
- ▶ By monotonicity, $H_{\min}^{\text{free}}(B_1)_\omega - H_{\min}^{\text{free}}(B_1|A_1)_{\rho \otimes \omega}$, as a function of ρ^{A_1} , is a **resource measure** for any ω^{B_1} .
 - non-increasing under free operations
 - equal to 0 if ρ^{A_1} is a free state
- ▶ A well-motivated resource measure of ρ^{A_1} :

$$\tilde{I}_{\min}^{\text{free}}(\rho) := \sup_{\omega^{B_1}} \left(H_{\min}^{\text{free}}(B_1)_\omega - H_{\min}^{\text{free}}(B_1|A_1)_{\rho \otimes \omega} \right)$$

Resource robustness as a reflection of informativeness

- ▶ Recall: $H_{\min}(B_1)_\omega \leq H_{\min}^{\text{free}}(B_1|A_1)_{\rho \otimes \omega} \leq H_{\min}^{\text{free}}(B_1)_\omega$ for ρ^{A_1} and ω^{B_1} .
- ▶ By monotonicity, $H_{\min}^{\text{free}}(B_1)_\omega - H_{\min}^{\text{free}}(B_1|A_1)_{\rho \otimes \omega}$, as a function of ρ^{A_1} , is a **resource measure** for any ω^{B_1} .
 - non-increasing under free operations
 - equal to 0 if ρ^{A_1} is a free state
- ▶ A well-motivated resource measure of ρ^{A_1} :

$$\tilde{I}_{\min}^{\text{free}}(\rho) := \sup_{\omega^{B_1}} \left(H_{\min}^{\text{free}}(B_1)_\omega - H_{\min}^{\text{free}}(B_1|A_1)_{\rho \otimes \omega} \right)$$

- how “informative” ρ^{A_1} could possibly be about an independent system B_1

Resource robustness as a reflection of informativeness

- ▶ Recall: $H_{\min}(B_1)_\omega \leq H_{\min}^{\text{free}}(B_1|A_1)_{\rho \otimes \omega} \leq H_{\min}^{\text{free}}(B_1)_\omega$ for ρ^{A_1} and ω^{B_1} .
- ▶ By monotonicity, $H_{\min}^{\text{free}}(B_1)_\omega - H_{\min}^{\text{free}}(B_1|A_1)_{\rho \otimes \omega}$, as a function of ρ^{A_1} , is a **resource measure** for any ω^{B_1} .
 - non-increasing under free operations
 - equal to 0 if ρ^{A_1} is a free state
- ▶ A well-motivated resource measure of ρ^{A_1} :

$$\tilde{I}_{\min}^{\text{free}}(\rho) := \sup_{\omega^{B_1}} \left(H_{\min}^{\text{free}}(B_1)_\omega - H_{\min}^{\text{free}}(B_1|A_1)_{\rho \otimes \omega} \right)$$

- how “informative” ρ^{A_1} could possibly be about an independent system B_1

Theorem (Entropic characterization of resource robustness)

$$\tilde{I}_{\min}^{\text{free}}(\rho) = \log(1 + \text{Rob}(\rho)), \text{ where } \text{Rob}(\rho) := \min\{r: \frac{\rho + r\tau}{1+r} \in \mathcal{D}_{\text{free}}^{A_1}, \tau \in \mathcal{D}^{A_1}\}.$$

Resource robustness as a reflection of informativeness

- ▶ Recall: $H_{\min}(B_1)_\omega \leq H_{\min}^{\text{free}}(B_1|A_1)_{\rho \otimes \omega} \leq H_{\min}^{\text{free}}(B_1)_\omega$ for ρ^{A_1} and ω^{B_1} .
- ▶ By monotonicity, $H_{\min}^{\text{free}}(B_1)_\omega - H_{\min}^{\text{free}}(B_1|A_1)_{\rho \otimes \omega}$, as a function of ρ^{A_1} , is a **resource measure** for any ω^{B_1} .
 - non-increasing under free operations
 - equal to 0 if ρ^{A_1} is a free state
- ▶ A well-motivated resource measure of ρ^{A_1} :

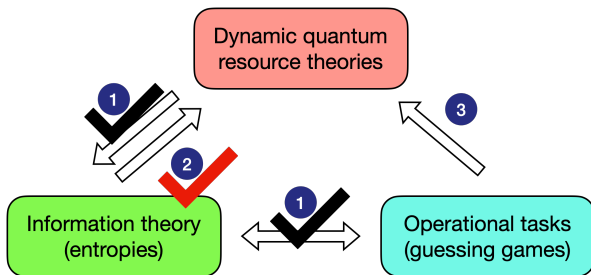
$$\tilde{I}_{\min}^{\text{free}}(\rho) := \sup_{\omega^{B_1}} \left(H_{\min}^{\text{free}}(B_1)_\omega - H_{\min}^{\text{free}}(B_1|A_1)_{\rho \otimes \omega} \right)$$

- how “informative” ρ^{A_1} could possibly be about an independent system B_1

Theorem (Entropic characterization of resource robustness)

$\tilde{I}_{\min}^{\text{free}}(\rho) = \log(1 + \text{Rob}(\rho))$, where $\text{Rob}(\rho) := \min\{r: \frac{\rho+r\tau}{1+r} \in \mathcal{D}_{\text{free}}^{A_1}, \tau \in \mathcal{D}^{A_1}\}$.

- ▶ Information-theoretic interpretation of resource robustness
- ▶ Similar result for channels



Generalizing entropies from quantum theory to quantum resource theories

Using free (conditional) min-entropy to characterize quantum resources

Characterizing dynamic quantum resources with operational tasks

Discussions

Characterizing dynamic quantum resources with operational tasks

- ▶ Entropic characterization:
 - relies on the **mathematical description** of the objects and resource theory
 - Useful quantities are evaluated by **computation**

Characterizing dynamic quantum resources with operational tasks

- ▶ Entropic characterization:
 - relies on the **mathematical description** of the objects and resource theory
 - Useful quantities are evaluated by **computation**
- ▶ What if:
 - ▶ Description of the state/channel is unknown
 - but copies of the state/channel can be obtained
 - ▶ Membership problem in the resource theory is computationally hard (e.g. deciding separability/entanglement is NP-hard)
 - but the experimenter is restricted by that resource theory in reality

Characterizing dynamic quantum resources with operational tasks

- ▶ Entropic characterization:
 - relies on the **mathematical description** of the objects and resource theory
 - Useful quantities are evaluated by **computation**
- ▶ What if:
 - ▶ Description of the state/channel is unknown
 - but copies of the state/channel can be obtained
 - ▶ Membership problem in the resource theory is computationally hard (e.g. deciding separability/entanglement is NP-hard)
 - but the experimenter is restricted by that resource theory in reality
- ▶ Operational characterization is more suitable:
 - relies on the **physical presence** of the objects and resource theory
 - Useful quantities are **observables** in operational tasks

Characterizing dynamic quantum resources with operational tasks

- ▶ The setting of an operational task:
 - ▶ A performer holding a channel resource Λ
 - ▶ A set of rules specified by parameters $\mathcal{G}$
 - ▶ A strategy space $\mathcal{S}_{\text{free}}$ for the performer, typically the set of free (super)operations
 - ▶ A reward function $r_i(\mathcal{G})$ conditioned on the outcome i

Characterizing dynamic quantum resources with operational tasks

- ▶ The setting of an operational task:
 - ▶ A performer holding a channel resource Λ
 - ▶ A set of rules specified by parameters $\mathcal{G}$
 - ▶ A strategy space $\mathcal{S}_{\text{free}}$ for the performer, typically the set of free (super)operations
 - ▶ A reward function $r_i(\mathcal{G})$ conditioned on the outcome i
- ▶ The performer's maximum expected reward when holding the resource Λ :

$$R^{\text{free}}(\Lambda; \mathcal{G}) = \max_{\Theta \in \mathcal{S}_{\text{free}}} \sum_i r_i(\mathcal{G}) p_i(\Theta[\Lambda]; \mathcal{G})$$

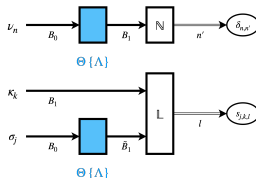
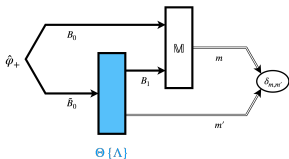
- ▶ $R^{\text{free}}(\Lambda; \mathcal{G})$, as a function of Λ , is a resource monotone for any $\mathcal{G}$.

Characterizing dynamic quantum resources with operational tasks

- ▶ The setting of an operational task:
 - ▶ A performer holding a channel resource Λ
 - ▶ A set of rules specified by parameters $\mathcal{G}$
 - ▶ A strategy space $\mathcal{S}_{\text{free}}$ for the performer, typically the set of free (super)operations
 - ▶ A reward function $r_i(\mathcal{G})$ conditioned on the outcome i
- ▶ The performer's maximum expected reward when holding the resource Λ :

$$R^{\text{free}}(\Lambda; \mathcal{G}) = \max_{\Theta \in S_{\text{free}}} \sum_i r_i(\mathcal{G}) p_i(\Theta[\Lambda]; \mathcal{G})$$

- ▶ $R^{\text{free}}(\Lambda; \mathcal{G})$, as a function of Λ , is a resource monotone for any $\mathcal{G}$.



Characterizing dynamic quantum resources with operational tasks

Theorem (Operational conditions for free convertibility between channels, informal)

For any channels $\Lambda^{A_0 \rightarrow A_1}$ and $\Psi^{B_0 \rightarrow B_1}$,

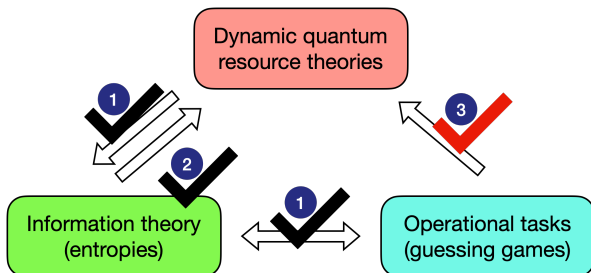
$$\Lambda \succsim_{\text{free}} \Psi \iff R^{\text{free}}(\Lambda; \mathcal{G}) \geq R^{\text{free}}(\Psi; \mathcal{G}) \quad \forall \mathcal{G} \in \mathcal{G}.$$

($\mathcal{G}$: a certain parameter set for a certain task).

Theorem (Operational characterization of resource robustness, informal)

$$\sup_{\mathcal{G} \in \mathcal{G}} \frac{R^{\text{free}}(\Lambda; \mathcal{G})}{\sup_{\Psi \in C_{\text{free}}} R^{\text{free}}(\Psi; \mathcal{G})} = 1 + \text{Rob}(\Lambda).$$

- ▶ Operational interpretation of resource robustness
- ▶ The result can be generalized to a broader family of robustness measures



Generalizing entropies from quantum theory to quantum resource theories

Using free (conditional) min-entropy to characterize quantum resources

Characterizing dynamic quantum resources with operational tasks

Discussions

Discussions

Outlook:

- ▶ Understanding uncertainty quantification in operationally restricted scenarios
 - ▶ Free min-entropy: formulation based on divergence; other entropic properties
 - ▶ Free other-entropy?
- ▶ Other operational approaches to generalizing entropy, e.g., based on the classical entropy of the measurement outcome ⁵
 - ▶ What's the connection?

⁵A. J. Short and S. Wehner, New J. Phys. 12 033023 (2010)